REST API
for
Web Service

REST API

- REST (**RE**presentation **S**tate **T**ransfer) API (**A**pplication **P**rogramming **I**nterface) has been used as standard web protocols to exchange data over the internet.
- Modern web services adopt it because it is lightweight, fast, and highly scalable.
- How it works:
  - Client sends an **HTTP Request** to Web Service address (called “Endpoint”)
  - Server processes the request then sends back **HTTP Response**, usually in the form of JSON.

Web Server VS Web Service

- Web Server responds to client (Web Browser) requests with HTML Pages

- Web Service responds to client (Browser, Mobile App, Desktop App) requests with XML or JSON

Traditional Web Request





Client

(Get/Post)

Request

Response

(HTML Page)



Server

Web Service (REST)



(Get/Post/Put/Delete)

Request





Response

(XML / JSON)

Client

Server

4 Main REST Methods (CRUD Operations)

- **REST APIs** use standard HTTP methods to perform the actions that match **CRUD** (Create Read Update Delete) operations.

HTTP Method	Action
GET	Read (Retrieve) Data from server resource
POST	Create New Data for server resource
PUT	Update Existing Data in server resource
DELETE	Delete Existing Data in server resource

Introduction to JSON-Server

JSON-Server

- JSON-Server provides simple yet powerful REST API
- Database’s tables (Resource) are stored in a JSON file.
- Client can perform CRUD on the resources provided by the Server

Setup std-phones API Server

- Create a folder “phone-server”
- Run the following scripts:
  - npm init
  - npm install json-server
  - npm install nodemon
- Edit file package.json
  - Edit “start” script as follow:

```
  "scripts": {
    "start": "nodemon --exec json-server -p 3008 phones.json"
  },
```

Student Phones Table

- Create a JSON file, “phones.json”

```
{
  "phones": [],
}
```

Table “phones” with no data

```
{
  "phones": [
    {
      "name": "Steve Davis",
      "sect": "CED",
      "tel": "088-886-9845",
      "id": "MfxT-wU6kf4"
    },
  ],
}
```

Table “phones” with a sample record.

Running Server

- Run JSON-Server script with node monitor:
  - npm start
  - The service will be run on the specified port (3008)

```
Endpoints:
http://localhost:3008/phones
```

Get Data resource from Endpoint

-



- npm install axios

```
import axios from 'axios'; // 1. Create the custom instance

const api = axios.create({
        baseURL: 'https://example.com',
        timeout: 5000, // 5 seconds
        headers: { 'Content-Type': 'application/json',
                'Authorization': 'Bearer YOUR_TOKEN_HERE' }
});
```

Axios Methods

- Get method

```
const response = await axios.get("https://example.com/resource");
console.log(response.data);
```

- Post method

```
const response = await axios.post(
    "https://example.com/resource",
     { data in JSON Object }
  );
  //return response.data;
```

Axios Method

- Put method

```
const response = await axios.put(
    "https://example.com/resource/id",
    { data in JSON Object }
 );
 //return response.data;
```

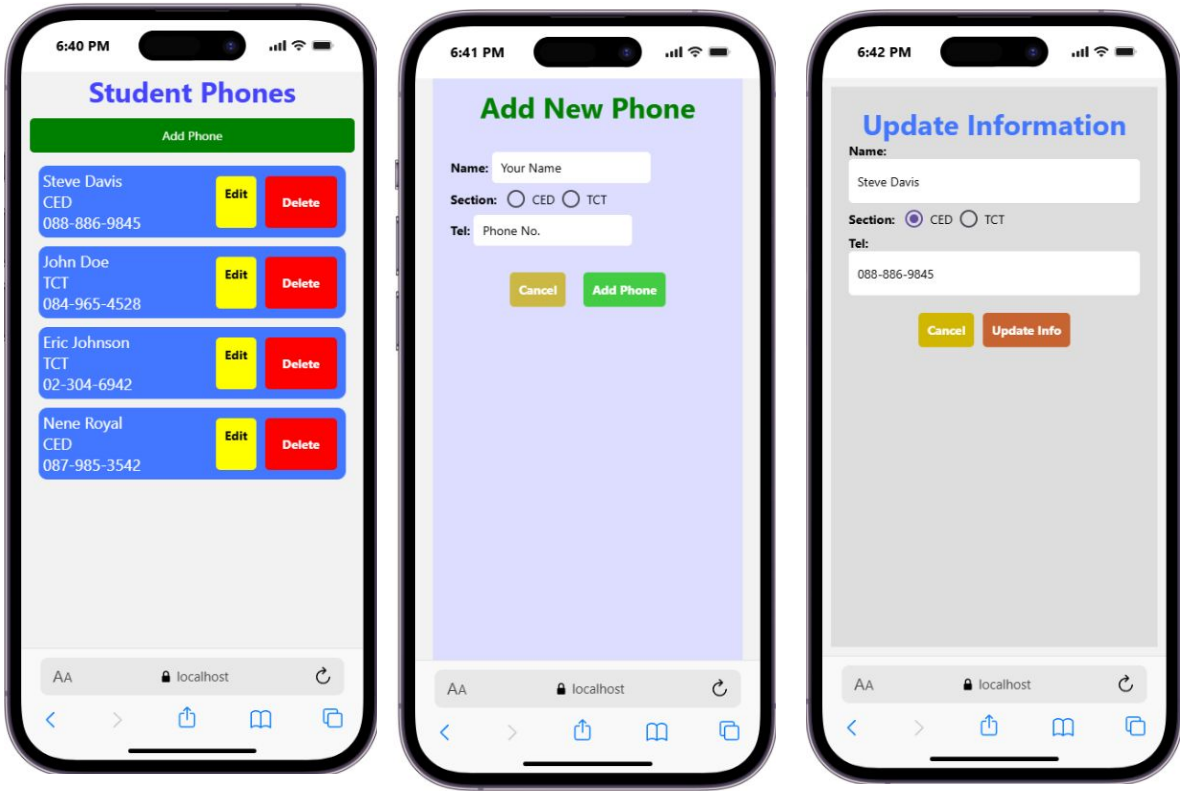
- Delete method

```
const response = await axios.delete(
    "https://example.com/resource/id"
 );
 //return response.data;
```

Student Phones App

Screen Shots

- Main Screen
- Add Screen
- Edit Screen



File Structure

- Install dependencies:
  - npm install uuid
  - npm install react-native-paper
  - npm install axios



_layout.tsx

- ```tsx
  import { Stack } from "expo-router";

  export default function RootLayout() {
    return (
      <Stack screenOptions={{ headerShown: false }}>
        <Stack.Screen name='index' />
        <Stack.Screen name='addPhone' />
        <Stack.Screen name='editPhone' />
      </Stack>
    );
  }
  ```
  ```

utils/crud-api.ts

- ```
  import axios from "axios";

  const api = axios.create({
    baseURL: "http://localhost:3008",
    headers: {
      "Content-Type": "application/json",
    },
  });

  export default api;
  ```
  ```

addPhone.tsx (1)

```
import api from '@/utils/crud-api';
import { useRouter } from 'expo-router';
import { useState } from 'react';
import { Alert, StyleSheet, Text, TextInput,
            TouchableOpacity, View } from 'react-native';
import { RadioButton } from 'react-native-paper'; // npm install ....
import { v4 as uuidv4 } from 'uuid';

export default function AddPhone() {
    const router = useRouter();
    const [name, setName] = useState('');
    const [sect, setSect] = useState('');
    const [tel, setTel] = useState('');
```

addPhone.tsx (2)

-

```
const addPhone = async () => {
    if(name===''|| sect==='' || tel===''){
        Alert.alert("Please Enter phone info");
        return;
    }
    let phone = {
        id: uuidv4(),
        name: name,
        sect: sect,
        tel: tel,
      }
    try {
        const res = await api.post('phones', phone);
        setName(''); setSect(''); setTel('');
        router.navigate('/');
    } catch(err) {
        console.log(err);
    }
}
```

addPhone.tsx (3)

-

```
    return(
        <View style={styles.form}>
            <Text style={styles.title}>Add New Phone</Text>
            <View style={{flexDirection: 'row', alignItems: 'center'}}>
                <Text style={{fontWeight: 'bold'}}>Name: </Text>
                <TextInput style={styles.input}
                        value={name}
                        onChangeText={(text)=> setName(text)}
                        placeholder='Your Name' />
            </View>
```

addPhone.tsx (4)

```
<RadioButton.Group  value={sect}
    onValueChange={value => setSect(value)}>
    <View style={{flexDirection: 'row', alignItems: 'center'}}>
        <Text style={{fontWeight: 'bold'}}>Section: </Text>
        <RadioButton value="CED" />
        <Text>CED</Text>
        <RadioButton value="TCT" />
        <Text>TCT</Text>
    </View>
</RadioButton.Group>
```

addPhone.tsx (5)

-

```
            <View style={{flexDirection: 'row', alignItems: 'center'}}>
                <Text style={{
                    fontWeight: 'bold',
                }}>Tel: </Text>
                <TextInput style={styles.input}
                        value={tel}
                        onChangeText={(text)=> setTel(text)}
                        placeholder='Phone No.' />
            </View>
```

addPhone.tsx (6)

```
            <View style={{flexDirection: 'row', justifyContent: 'center', marginTop: 20,}}>
            <TouchableOpacity onPress={()=>router.back()}
                style={{backgroundColor: 'rgb(204, 184, 68)', padding: 10,
                    margin: 10, borderRadius: 5,}}>
                <Text style={{color: 'white', fontWeight: 'bold',
                    textAlign: 'center',
                }}>Cancel</Text>
            </TouchableOpacity>
            <TouchableOpacity onPress={()=>addPhone()}
                style={{backgroundColor: '#4C4', padding: 10,
                    margin: 10, borderRadius: 5,}}>
                <Text style={{color: 'white', fontWeight: 'bold',
                    textAlign: 'center',
                }}>Add Phone</Text>
            </TouchableOpacity>
            </View>
        </View>
    )
}
```

addPhone.tsx (7)

-

```
const styles = StyleSheet.create({
    title: {
        textAlign: 'center',
        fontSize: 32,
        fontWeight: 'bold',
        color:'green',
        marginTop: 10,
        marginBottom: 30,
    },
```

addPhone.tsx (8)

-

```
    form: {
        flex: 1,
        backgroundColor: '#DDF',
        paddingHorizontal: 20,
        marginHorizontal: 20,
    },
    input: {
        height: 35,
        borderColor: '#ccc',
        padding: 10,
        borderRadius: 5,
        backgroundColor: 'white',
    },
})
```

editPhone.tsx (1)

```
import { useLocalSearchParams, useRouter } from 'expo-router';
import { useState } from 'react';
import { Alert, StyleSheet, Text,
         TextInput, TouchableOpacity, View } from 'react-native';
import { RadioButton } from 'react-native-paper'; // npm install ....
import api from '../utils/crud-api';

const EditPhone = () => {
    const {id, name, sect, tel} = useLocalSearchParams();
    const [newName, setNewName] = useState(name);
    const [newSect, setNewSect] = useState(sect);
    const [newTel, setNewTel] = useState(tel);
    const router = useRouter();
```

editPhone.tsx (2)

-

```
    const updatePhone = async () => {
        if(name===''|| sect==='' || tel===''){
            console.log("Please Enter phone info");
            Alert.alert("Please Enter phone info");
            return;
        }
        try {
            const res = await api.put('phones/'+id, {
                name: newName,
                sect: newSect,
                tel: newTel
            });
            router.navigate('/');
            //console.log(res);
        } catch(err) {
            console.log(err);
        }
    }
```

editPhone.tsx (3)

```
    return(
        <View style={styles.container}>
            <Text style={styles.title}>Update Information</Text>
            <Text style={{fontWeight: 'bold'}}>Name: </Text>
            <TextInput style={styles.input}
                    value={newName}
                    onChangeText={(text)=> setNewName(text)}
                    placeholder='Your Name' />
            <RadioButton.Group  value={newSect}
                onValueChange={value => setNewSect(value)}>
                <View style={{flexDirection: 'row', alignItems: 'center'}}>
                    <Text style={{fontWeight: 'bold'}}>Section: </Text>
                    <RadioButton value="CED" />
                    <Text>CED</Text>
                    <RadioButton value="TCT" />
                    <Text>TCT</Text>
                </View>
            </RadioButton.Group>
```

editPhone.tsx (4)

```
            <Text style={{fontWeight: 'bold'}}>Tel: </Text>
            <TextInput style={styles.input}
                    value={newTel}
                    onChangeText={(text)=> setNewTel(text)}
                    placeholder='Phone No.' />

            <View style={{flexDirection: 'row',
                        justifyContent: 'center', marginTop: 20,}}>
                <TouchableOpacity onPress={()=>router.back()}
                    style={{backgroundColor: 'rgb(209, 182, 2)', padding: 10,
                        borderRadius: 5, marginRight: 10,
                    }}>
                    <Text style={{color: 'white', fontWeight: 'bold',
                        textAlign: 'center',
                    }}>Cancel</Text>
                </TouchableOpacity>
```

editPhone.tsx (5)

```
                <TouchableOpacity onPress={()=>updatePhone()}
                    style={{backgroundColor: 'rgb(200, 100, 50)', padding: 10,
                        borderRadius: 5,
                    }}>
                    <Text style={{color: 'white', fontWeight: 'bold',
                        textAlign: 'center',
                    }}>Update Info</Text>
                </TouchableOpacity>
            </View>
        </View>
    )
}

export default EditPhone;
```

editPhone.tsx (6)

- ```
const styles = StyleSheet.create({
    container: {
        flex: 1,
        margin: 10,
        padding: 20,
        backgroundColor: '#ddd',
    },
    title:{
        fontSize: 32,
        color: '#47F',
        fontWeight: 'bold',
        textAlign: 'center',
    },
```
```

editPhone.tsx (7)

- ```
    form: {
        flexDirection: 'row',
        alignItems: 'center',
        justifyContent: 'flex-end',
        backgroundColor: '#DDF',
        paddingHorizontal: 20,
    },
    input: {
        height: 50,
        borderColor: '#ccc',
        padding: 10,
        borderRadius: 5,
        backgroundColor: 'white',
    },
});
```
```

index.tsx (1)

```tsx
import Card from "@/components/card";
import api from "@/utils/crud-api";
import { router } from "expo-router";
import { useEffect, useState } from "react";
import { FlatList, Pressable, StyleSheet, Text, View } from "react-native";

export default function Index() {
  const [data, setData] = useState([]);

  useEffect(()=>{
      getData();
  }, [])
```

index.tsx (2)

-

```
const getData = async () => {
  try {
    const response = await api.get("phones");
    setData(response.data);
    //console.log(response.data);
  } catch(err) {
    console.log("ERROR", err);
  }

}
```

index.tsx (3)

-

```
  return (
    <View style={styles.container}>
      <Text style={styles.title}>Student Phones</Text>
      <Pressable onPress={()=>router.push('/addPhone')}
        style={{backgroundColor: 'green', padding: 10,
          margin: 10, borderRadius: 5,}}>
        <Text style={{color: 'white', textAlign: 'center'}}>Add Phone</Text>
      </Pressable>
```

index.tsx (4)

-

```
      <FlatList
        data={data}
        keyExtractor={item => item.id}
        renderItem={({item}) => {
          let phone = {
            id: item.id,
            name: item.name,
            sect: item.sect,
            tel: item.tel,
          }
          return (
            <Card phone={phone} refresh={getData} />
          )
        }}
      />
    </View>
  );
```

index.tsx (5)

-

```
const styles = StyleSheet.create({
  container: {
    flex: 1,
    justifyContent: "center",
  },
  title: {
    textAlign: 'center',
    fontSize: 32,
    fontWeight: 'bold',
    color: '#44F',
  },
});
```

components/card.tsx (1)

```tsx
import { Link } from "expo-router";
import { StyleSheet, Text, TouchableOpacity, View } from "react-native";
import api from "../utils/crud-api";

export default function Card(props) {

  const delPhone = async (id: string) => {
    console.log("Delete phone with id:", id);
        try {
            const res = await api.delete('phones/' + id);
            props.refresh();
            console.log(res);
        } catch(err) {
            console.log(err);
        }
```

components/card.tsx (2)

```
/*
    Alert.alert(
      "Are you sure?", // Title
      "Do you really want to delete this file?", // Message
      [
        {
          //text: "Cancel",
          //onPress: () => console.log("Cancel Pressed"),
          //style: "cancel" // Applies default cancel styling (iOS only)
        },
        {
          text: "Delete",
          onPress: () => console.log("Delete Pressed"),
          style: "destructive" // Applies red text styling (iOS only)
        }
      ],
      { cancelable: true } // Allows closing the popup by tapping outside (Android only)
    );
    */
  };
```

components/card.tsx (3)

```
    return (
        <View style={styles.container} key={props.phone.id}>
          <View style={styles.text}>
            <Text style={styles.text}>{props.phone.name}</Text>
            <Text style={styles.text}>{props.phone.sect}</Text>
            <Text style={styles.text}>Tel No.: {props.phone.tel}</Text>
          </View>
          <View style={{flexDirection: 'row'}}>
              <Link href={{
                pathname: "/editPhone",
                params: {
                    id: props.phone.id,
                    name:  props.phone.name,
                    sect: props.phone.sect,
                    tel: props.phone.tel,
                  },
                }}
```

components/card.tsx (4)

-

```
                push style={{backgroundColor: 'yellow', padding: 10,
                    margin: 5, borderRadius: 5,}}>
                    <Text style={{
                      alignItems: 'center',
                      justifyContent: 'center',
                      color: 'black',
                      fontWeight: 'bold',
                      textAlign: 'center',
                    }}>Edit</Text>
            </Link>
```

components/card.tsx (5)

```
            <TouchableOpacity onPress={()=>delPhone(props.phone.id)}
               style={{backgroundColor: 'red', padding: 10,
                    margin: 5, borderRadius: 5,}}>
                <Text
                  style={{padding: 5, margin: 5, borderRadius: 5,
                               color: 'white', fontWeight: 'bold',
                               textAlign: 'center',}}>
                    Delete
                </Text>
            </TouchableOpacity>
          </View>
        </View>
    )
}
```

components/card.tsx (6)

-

```
const styles = StyleSheet.create({
  container: {
    flex: 1,
    flexDirection: 'row',
    alignItems: "center",
    justifyContent:"space-between",
    borderRadius: 10,
    padding: 5,
    marginVertical: 5,
    marginHorizontal: 20,
    backgroundColor: '#47F',
  },
```

components/card.tsx (7)

-

```
  list: {
    backgroundColor: '#47F',
    padding: 10,
    margin: 5,
    borderRadius: 10,
    borderColor: 'red',
    alignItems: 'center',
  },
  text: {
    color: 'white',
    fontSize: 18,
  }
});
```